

PEC Viva & Interview Defense Guide – Cyber Security

Common Viva Questions and Model Answers

Question: What is CIA Triad?

Answer: The CIA Triad consists of Confidentiality, Integrity, and Availability, which form the foundation of information security. Confidentiality ensures data is accessible only to authorized users, Integrity maintains data accuracy and consistency, and Availability ensures systems and data are accessible when needed.

Question: Explain SIEM.

Answer: SIEM (Security Information and Event Management) is a comprehensive security solution that aggregates logs from various sources, correlates events in real-time, and provides actionable security alerts. It helps detect threats, ensure compliance, and investigate security incidents.

Question: Difference between IDS and IPS?

Answer: IDS (Intrusion Detection System) passively monitors network traffic and detects potential attacks by alerting administrators. IPS (Intrusion Prevention System) actively detects and blocks malicious traffic in real-time, preventing attacks from reaching their targets.

Question: What is Zero Trust Security?

Answer: Zero Trust is a security model based on the principle 'never trust, always verify.' It assumes no user or device should be trusted by default, requiring continuous authentication and authorization for every access request, regardless of location.

Question: Explain Defense in Depth.

Answer: Defense in Depth is a layered security strategy that uses multiple security controls at different levels (network, application, data, physical). If one layer fails, others continue to provide protection, reducing the risk of successful attacks.

Question: What is a firewall?

Answer: A firewall is a network security device that monitors and controls incoming and outgoing traffic based on predetermined security rules. It acts as a barrier between trusted internal networks and untrusted external networks.

Question: What is Penetration Testing?

Answer: Penetration testing is an authorized simulated cyberattack on a system to identify vulnerabilities that could be exploited by malicious actors. It helps organizations strengthen their security posture before real attacks occur.

Question: Explain the concept of Least Privilege.

Answer: Least Privilege is a security principle where users and systems are granted only the minimum level of access rights needed to perform their functions. This limits potential damage from accidents, errors, or unauthorized access.

Question: What is Multi-Factor Authentication?

Answer: MFA is a security mechanism requiring users to provide two or more verification factors to gain access. This typically includes something you know (password), something you have (token), or something you are (biometric).

Question: What is the OWASP Top 10?

Answer: OWASP Top 10 is a regularly updated list of the most critical web application security risks, including injection attacks, broken authentication, sensitive data exposure, and security misconfigurations. It serves as a security awareness document for developers.

Interview Tips & Best Practices

- Stay calm and composed - take a moment to think before answering
- If you don't know an answer, be honest and show willingness to learn
- Use real-world examples when explaining concepts
- Demonstrate practical understanding, not just theoretical knowledge
- Ask for clarification if a question is unclear
- Relate your answers to current cybersecurity trends and incidents
- Show enthusiasm for the field and continuous learning mindset
- Prepare examples of projects or labs you've worked on
- Be ready to discuss latest security news and vulnerabilities
- Practice explaining technical concepts in simple terms